FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Diagnosis of Pregnancy & Physiological Changes

Lecture | Week 1 · Foundations, ANC & early pregnancy | Foundations | Prof Mirembe

Pregnancy is diagnosed from symptoms (amenorrhoea, nausea, breast changes), signs (uterine enlargement, Hegar's, Chadwick's, fetal heart) and tests (hCG, ultrasound). The mother's body then adapts in every system, including a 40–50% rise in plasma volume, a 30–50% rise in cardiac output, respiratory alkalosis, a 50% rise in GFR, a hypercoagulable state and insulin resistance. Knowing these normal changes lets you tell physiology from disease and explains why pregnant women bleed, clot, desaturate and decompensate differently.

Must know before the ward round

1. Urine hCG is positive from about the time of the missed period. Serum β-hCG is detectable 8–11 days after conception, doubles about every 48 hours early on and peaks at 8–10 weeks.
2. On transvaginal ultrasound a gestational sac is seen at about 4.5–5 weeks, a yolk sac at about 5.5 weeks and cardiac activity at about 6 weeks. The discriminatory hCG zone is about 1,500–3,500 IU/L.
3. Positive (certain) signs are fetal heart sounds, fetal parts or movements felt by the examiner, and the fetus seen on ultrasound. Everything else is presumptive or probable.
4. Plasma volume rises 40–50% but red cell mass only 20–30%, so there is physiological haemodilution. Hb below 11 g/dL is still anaemia.
5. Cardiac output rises 30–50% (stroke volume, then heart rate up by 10–20 bpm). SVR falls, so BP dips in the second trimester. The supine gravid uterus compresses the IVC from about 20 weeks, so tilt her left.
6. Minute ventilation rises about 40–50% through tidal volume (progesterone), $PaCO_2$ falls to about 4 kPa (30 mmHg) and FRC falls about 20%. A pregnant woman desaturates fast.
7. GFR and renal plasma flow rise about 50%, so a 'normal' creatinine of 90 µmol/L in pregnancy is actually abnormal.
8. Pregnancy is hypercoagulable (fibrinogen and factors VII, VIII and X up; protein S down), so VTE risk is about 5-fold, highest postpartum.
9. Do a urine hCG in every woman of reproductive age with abdominal pain, bleeding, collapse, or before a drug, X-ray or operation. Positive hCG + pain + shock means ruptured ectopic until proven otherwise.
10. Pregnancy normal ranges to quote: creatinine 35–70 µmol/L, $PaCO_2$ about 4 kPa, HCO_3^- 18–22 mmol/L, fibrinogen 4–6 g/L, ALP up 2–4 times, albumin 28–37 g/L.
11. Never physiological in pregnancy are RR above 20/min, SpO_2 below 95%, resting HR above 100, a diastolic murmur, raised ALT/AST and unilateral leg swelling.
12. A pregnant woman can lose up to about 30% of her blood volume before BP falls. Shock index (HR ÷ SBP) 0.9 or more is worrying and 1.4 or more is critical.

1 Definitions and Key Concepts
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3 Principles and Framework
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6 Common Problems and Their Management
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1. Definitions and Key Concepts

This lecture has two halves that belong together.

1. **Diagnosing pregnancy** is the first clinical decision in obstetrics. Every woman of reproductive age with abdominal pain, bleeding, collapse or who needs a drug or X-ray **is pregnant until proven otherwise**. In Uganda, a urine hCG test costs very little and prevents missed ectopic pregnancies, teratogenic prescriptions and wrong diagnoses.
2. **Physiological changes** are the body's preparation to grow a fetus and survive delivery: more blood, more cardiac output, more ventilation, more clotting, more insulin resistance. These changes are why pregnant women **tolerate haemorrhage until they suddenly crash**, why they **desaturate quickly** during eclampsia or anaesthesia, why **VTE** is a leading killer, and why normal laboratory values differ.

The one thing never to forget: **"normal" values change in pregnancy.** A creatinine, $PaCO_2$ or heart rate that is normal for a non-pregnant woman may be a warning sign in a pregnant one, and vice versa, since a lower Hb and higher WCC can be physiological.

Signs and symptoms of pregnancy: the three classic categories

Category	Examples	Why not certain?
Presumptive (mostly symptoms)	Amenorrhoea; nausea/vomiting ("morning sickness", from about 6 weeks, peaking at 9–12 weeks); breast tenderness and enlargement; urinary frequency; fatigue; quickening (the first fetal movements the mother feels, at **18–20 weeks in a primigravida** and **16–18 weeks in a multigravida**); skin changes (chloasma, linea nigra, striae)	Can occur with stress, hormonal contraception, thyroid disease, hyperprolactinaemia, pseudocyesis, bowel gas
Probable (signs found by the examiner)	Uterine enlargement; **Hegar's sign** (softening of the isthmus, 6–10 weeks); **Goodell's sign** (softening of the cervix, about 6–8 weeks); **Chadwick's sign** (bluish/purple discolouration of the vagina and cervix from congestion, 6–8 weeks); **Osiander's sign** (increased pulsation in the lateral fornices); ballottement (about 16–20 weeks); Braxton Hicks contractions; **positive hCG test**	Fibroids, molar pregnancy, pelvic congestion, choriocarcinoma or hCG-secreting tumours can mimic them
Positive (certain)	**Fetal heart** heard (Doppler from about 10–12 weeks; Pinard from about 20 weeks); **fetal parts or movements felt by the examiner** (about 20 weeks); **fetus seen on ultrasound**	These come only from the fetus

Dating terms

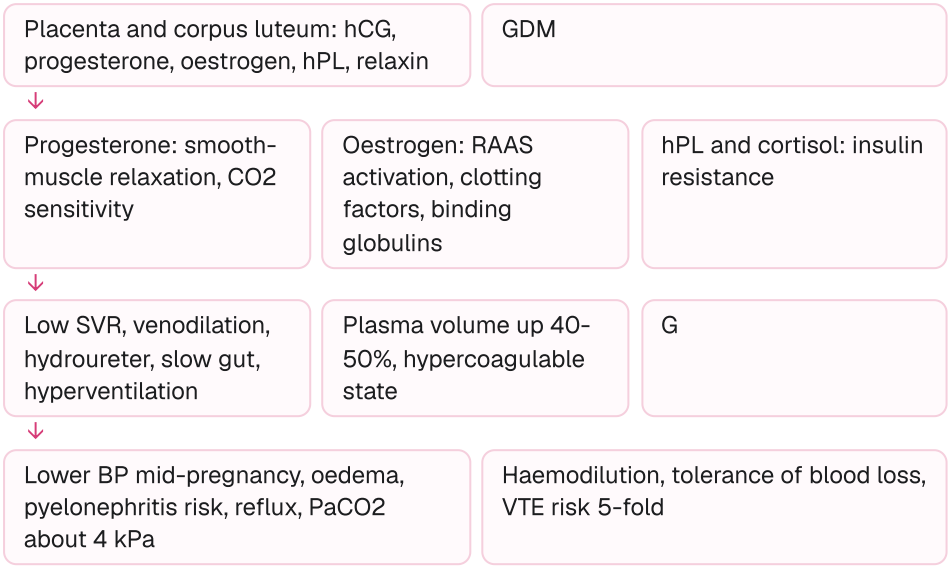
Term	Definition
Gestational age (GA)	Weeks from the first day of the LNMP. About 2 weeks more than the **conceptional (embryonic) age**
Naegele's rule	EDD = LNMP **+ 7 days – 3 months + 1 year** (the same as + 9 months + 7 days, or + 280 days). Adjust for a longer cycle: add (cycle length – 28) days
Trimesters	First up to 13+6 weeks; second 14–27+6 weeks; third 28 weeks to delivery
Embryonic vs fetal period	Embryo until 10 weeks' GA (8 weeks post-conception): organogenesis, the most teratogen-sensitive time. Fetus after that

2. Burden and Uganda Context

- Uganda has one of the highest fertility rates in the world (**about 4.5 children per woman**, UDHS 2022) and many unintended pregnancies, especially among adolescents (about 1 in 4 girls aged 15–19 has begun childbearing).
- **Late recognition of pregnancy** drives late ANC booking. Many women attend first in the second trimester, which means late dating, missed early folate and missed early risk assessment.
- **Ectopic pregnancy** is a common surgical emergency in Ugandan hospitals and often presents late, ruptured, with haemoperitoneum. A negative-until-proven urine hCG in any woman with lower abdominal pain is lifesaving.
- **Ultrasound** is widely available in towns (private clinics, HC IV and hospitals) but less so rurally. Clinical skills (bimanual size, fundal height, fetal heart with a Pinard or Doppler) still matter.
- Background anaemia, malaria, HIV and rheumatic heart disease interact with the physiological changes. A woman with **mitral stenosis** may first decompensate at **28–32 weeks** (when cardiac output nears its peak) or **immediately after delivery** (autotransfusion).

3. Principles and Framework

The physiological changes follow from a few hormones (table below) acting on every system. Learn each change with its **reason** and its **clinical consequence**: that is how they are examined and how you will use them on the ward.



Hormones driving the changes

Hormone	Source	Key effects
hCG	Syncytiotrophoblast	Keeps the **corpus luteum** alive (progesterone until the placenta takes over at about 8–10 weeks); peaks at 8–10 weeks (about 100,000 IU/L), then falls to a plateau. Has **TSH-like activity** (transient low TSH, hyperemesis). Very high in **molar pregnancy and twins**
Progesterone	Corpus luteum, then placenta	**Smooth-muscle relaxation** (GI, ureters, veins, uterus, the "pro-gestation" hormone that keeps the uterus quiet); raises the respiratory centre's sensitivity to CO_2 (hyperventilation); raises body temperature by about 0.5 °C
Oestrogens (mostly oestriol)	Placenta (with fetal adrenal and liver precursors)	Uterine growth and blood flow; breast ductal growth; **raises clotting factors, TBG and cortisol-binding globulin**; skin changes

Hormone	Source	Key effects
hPL (human placental lactogen)	Placenta	**Insulin resistance** (diverts glucose to the fetus), lipolysis
Relaxin	Corpus luteum, placenta	Ligament laxity (pelvic joints, symphysis), cervical remodelling
Prolactin	Anterior pituitary (which enlarges)	Prepares the breast for lactation (milk is held back by high progesterone until the placenta is delivered)

Cardiovascular system

Parameter	Change	Why / clinical relevance
Cardiac output	**Up 30–50%**, starting by 5–8 weeks and near peak by about 24–28 weeks	Stroke volume up first, then heart rate. Further **up to 60–80% in labour and immediately after delivery** (autotransfusion of about 500 mL from the contracting uterus), which is the danger time for cardiac disease
Heart rate	**Up 10–20 bpm**	A resting HR above 100 bpm is not normal and needs explaining (bleeding, sepsis, anaemia, PE, thyroid disease)
Stroke volume	Up 20–30%	Increased preload from volume expansion
Systemic vascular resistance	**Down about 20–30%**	Progesterone, nitric oxide, and a low-resistance placental bed
Blood pressure	Falls in the first and second trimesters (**nadir at about 20–24 weeks**, diastolic falls most), returns to baseline by term	Hypertension before 20 weeks suggests **chronic hypertension** (or molar pregnancy); physiological lowering can mask it
Heart position and signs	Heart pushed up and left (apex displaced); **ejection systolic flow murmur** in about 90%; third heart sound common	**Diastolic murmurs are never physiological**, so investigate them (mitral stenosis is common in Uganda)
ECG	Left axis deviation, T-wave inversion in III, ectopics	Do not over-read it
Venous pressure in the legs	Rises (uterine compression of the iliac veins and IVC)	Ankle oedema (about 80% of normal pregnancies), varicose veins, haemorrhoids, **DVT (commoner on the left)**

CLINICAL PEARL

Aortocaval compression. From about **20 weeks** the supine uterus compresses the IVC (and aorta), reducing venous return and cardiac output by up to 25–30%. It causes **supine hypotensive syndrome** (dizziness, pallor, fetal distress). In resuscitation, CPR, examination and CS, **tilt her 15° to the left or manually displace the uterus to the left**. In cardiac arrest after 20 weeks, if there is no response by **4 minutes**, a perimortem CS (resuscitative hysterotomy) is done to save the **mother**.

Blood and haematology

Parameter	Change	Clinical relevance
Plasma volume	**Up 40–50%** (about 1,200–1,500 mL), mostly by 32–34 weeks	Protects against the blood loss of delivery. Driven by RAAS activation (oestrogen raises angiotensinogen) and water retention (osmotic threshold resets, with sodium about 4–5 mmol/L lower)
Red cell mass	**Up 20–30%** (more with iron supplements)	Rises less than plasma, giving **physiological (dilutional) anaemia**, with the lowest Hb at about 28–34 weeks
Haemoglobin	Falls (normal lower limit **11 g/dL** in pregnancy, WHO)	Below 11 is anaemia. **Poor plasma expansion** (high Hb/haematocrit) is a sign of **pre-eclampsia**
Iron requirement	About **1 g** in total over the pregnancy (fetus and placenta about 300 mg, red cell expansion about 450–500 mg, losses about 200 mg)	Most women cannot meet this from diet, hence **routine iron**
White cells	Up (up to about 15 × 10^9/L; **up to 25 × 10^9/L in labour and the early puerperium**)	A raised WCC alone does not prove infection
Platelets	Slight fall (dilution and consumption). **Gestational thrombocytopenia**: mild (usually above 70–100 × 10^9/L), third trimester, asymptomatic	Platelets below 100 × 10^9/L with hypertension suggests **HELLP**
Clotting factors	**Fibrinogen up** (to about 4–6 g/L), factors **VII, VIII, IX, X and vWF up**; **protein S down**, **activated protein C resistance**; fibrinolysis **reduced** (PAI-1 and PAI-2 up)	**Hypercoagulable state** prepares for placental separation but gives about a **5-fold VTE risk**, highest **postpartum** (especially after CS). A "normal" fibrinogen of 2 g/L in a bleeding woman is actually low, which suggests **DIC**
ESR	Rises markedly	Useless as a marker of inflammation in pregnancy

Virchow's triad in pregnancy: stasis (venous compression, progesterone venodilation, immobility), hypercoagulability (the factors above), and vessel wall injury (delivery, CS).

Respiratory system

Parameter	Change	Clinical relevance
Tidal volume	**Up 30–50%**	Progesterone sensitises the respiratory centre to CO_2
Respiratory rate	**Unchanged** (or minimally up)	**RR above 20/min is abnormal**: think PE, pneumonia, pulmonary oedema, sepsis, DKA
Minute ventilation	**Up about 40–50%**	"Physiological dyspnoea of pregnancy" (breathless feeling, normal exam, normal SpO_2)
Oxygen consumption	Up about 20%	Fetus, placenta, maternal organs
FRC (functional residual capacity)	**Down about 20%** (diaphragm elevated by up to 4 cm)	Smaller O_2 reserve plus higher consumption means **rapid desaturation** in apnoea (eclamptic fit, intubation)

Parameter	Change	Clinical relevance
Arterial blood gas	**$PaCO_2$ falls to about 4.0 kPa (30 mmHg)**; pH about 7.42–7.44; **HCO_3^- falls to about 18–22 mmol/L** (renal compensation); PaO_2 slightly up	**Compensated respiratory alkalosis.** The low $PaCO_2$ gradient helps the fetus offload CO_2. A "normal" $PaCO_2$ of 5.3 kPa (40 mmHg) in a pregnant asthmatic suggests **tiring and impending respiratory failure**
Upper airway	Oedema and hyperaemia of the mucosa	Nasal congestion and nosebleeds; **difficult intubation** (use a smaller tube)
Vital capacity	Unchanged	Chest widens (subcostal angle increases) to compensate for the elevated diaphragm

Renal and urinary tract

Parameter	Change	Clinical relevance
Renal plasma flow	Up 50–80%	–
GFR	**Up about 50%** (from early pregnancy)	**Creatinine falls** (normal about 35–70 µmol/L; above about 77–80 µmol/L is suspicious); **urea falls**. Drugs cleared renally (e.g. some antibiotics, $MgSO_4$) are cleared faster
Glucose handling	Filtered load exceeds tubular reabsorption, so **glycosuria** is common	Glycosuria alone does not diagnose diabetes; screen with blood glucose/OGTT
Protein	Slight rise in excretion (normal below 300 mg/24 h)	Proteinuria of 300 mg/24 h or more (or PCR 30 mg/mmol or more) is abnormal
Uric acid	Falls early (GFR) then rises to non-pregnant levels at term	A **high urate** is associated with pre-eclampsia
Ureters and pelvis	**Dilated (hydroureter/hydronephrosis), right more than left**, from progesterone relaxation plus compression at the pelvic brim (the right ovarian vein and dextrorotation of the uterus)	Urinary stasis, so **asymptomatic bacteriuria progresses to pyelonephritis** (about 30% if untreated). Treat bacteriuria. Do not misdiagnose obstruction on ultrasound
Bladder	Frequency (early from uterine pressure, late from the head in the pelvis)	–

Gastrointestinal system and liver

- **Nausea and vomiting** (linked to hCG and oestrogen): affects about 70–80% of women, usually settles by 14–16 weeks. **Hyperemesis gravidarum** means dehydration, ketosis and more than 5% weight loss.
- **Progesterone relaxes smooth muscle**, which causes **heartburn/reflux** (lower oesophageal sphincter), delayed gastric emptying (**aspiration risk under general anaesthesia**: give an antacid and use a rapid-sequence induction), **constipation**, and **gallbladder stasis** (gallstones).
- **Liver:** size and blood flow essentially unchanged. **ALP rises 2–4 times** (placental isoenzyme, so it is normal). **Albumin falls** (dilution). Transaminases (ALT, AST), bilirubin and GGT are normal or slightly lower, so **raised ALT/AST in pregnancy is always abnormal** (HELLP, acute fatty liver, intrahepatic cholestasis, hepatitis).
- **Gums:** hyperaemia and bleeding (pregnancy gingivitis, epulis).
- **Pica** (eating soil or clay, "emumbwa") is common in Uganda and linked with iron deficiency and helminths.

Endocrine and metabolic

System	Change	Relevance
Carbohydrate	First half: insulin sensitivity is normal or increased (a tendency to **hypoglycaemia** with fasting). Second half: **insulin resistance** (hPL, cortisol, progesterone, prolactin) diverts glucose to the fetus	This unmasks **GDM**, so screen at **24–28 weeks**
Lipids	Triglycerides and cholesterol up	Energy store for late pregnancy and lactation
Thyroid	Gland enlarges modestly; **TBG up**, so **total** T4/T3 are up but **free** T4 is normal; **TSH falls in the first trimester** (hCG stimulates the TSH receptor)	Use trimester-specific ranges. Iodine needs rise. A goitre should still be examined
Adrenal	Cortisol (total and free) up; aldosterone, renin and angiotensin up	Striae, glucose intolerance; sodium and water retention
Pituitary	Enlarges (up to 2-fold, lactotroph hyperplasia); prolactin rises	**Sheehan's syndrome** (postpartum pituitary necrosis after massive PPH): failure of lactation, amenorrhoea
Calcium	Total calcium falls (albumin), ionised calcium normal; **1,25-$(OH)_2$ vitamin D doubles** so gut absorption doubles	The fetus takes about 30 g of calcium, mostly in the third trimester

Reproductive organs

- **Uterus:** from about 50–70 g and 10 mL capacity to about **1,000–1,100 g** and **about 5 L** at term. Growth is by **hypertrophy** (and early hyperplasia) of myometrial cells. **Uterine blood flow** rises from about 50 mL/min to about **500–750 mL/min at term** (about 10–15% of cardiac output), which is why **PPH can exsanguinate a woman within minutes**. The **lower segment** forms from the isthmus from about 28 weeks, is thin and less vascular, and is where we make the CS incision.
- **Fundal landmarks:** at 12 weeks the uterus becomes palpable just above the symphysis; at **16 weeks** it lies midway between the symphysis and umbilicus; at **20–22 weeks** at the umbilicus; at **36 weeks** at the xiphisternum; at 40 weeks it drops slightly with **lightening** (engagement in primigravidae).
- **Cervix:** softens (Goodell's) and becomes vascular (bluish). Glands proliferate to form the **mucus plug (operculum)**. Physiological **ectropion** (contact bleeding). Remodels near term (ripening).
- **Vagina:** congested (Chadwick's), increased discharge (**leucorrhoea**, white, non-offensive, **lactobacilli**, acidic), with a higher risk of **candidiasis**.
- **Ovaries:** ovulation stops. The **corpus luteum** supports the pregnancy until about 8–10 weeks. Removing it before then without progesterone support causes miscarriage.
- **Breasts:** enlarge, tender, **Montgomery's tubercles** (enlarged sebaceous glands of the areola), secondary areola, pigmented nipples, colostrum from about 16 weeks.

Skin and musculoskeletal

- Pigmentation from **MSH/oestrogen/progesterone**: **linea nigra**, **chloasma/melasma** ("mask of pregnancy"), darkened areolae.
- **Striae gravidarum**, spider naevi and palmar erythema (oestrogen).
- **Lordosis** and back pain; **symphysis pubis dysfunction** (relaxin); **carpal tunnel syndrome** (fluid retention).

Weight gain and nutrition

- Typical total **11–16 kg** for a woman of normal BMI (IOM). The components are fetus about 3.4 kg, placenta 0.6, amniotic fluid 0.8, uterus 0.9, breasts 0.4, blood 1.5, extracellular fluid 1.5 and fat about 3.5 kg.

- Energy needs rise by about 300 kcal/day in the second and third trimesters, so advise "**one extra meal a day**" in Uganda.

4. Application in Practice

This topic has two practical workups that you will use every day: **(A) confirming and dating a pregnancy**, and **(B) deciding whether a symptom, sign or result in a pregnant woman is normal physiology or disease**. Both follow the same steps.

Step 0: First 5 minutes

1. **Is she pregnant?** Any woman aged about 12–50 with abdominal pain, bleeding, collapse, vomiting, or who needs a drug, X-ray or operation gets a **urine hCG** now. It takes 5 minutes and costs very little.
2. **Vital signs, read with pregnancy norms:**

Sign	Normal in pregnancy	Pathological: act now
Heart rate	Up 10–20 bpm (roughly 80–95/min)	**Above 100/min at rest**: bleeding, sepsis, anaemia, PE, thyroid, cardiac disease
Blood pressure	Lower in the 2nd trimester (nadir 20–24 weeks)	**140/90 or more** = hypertension; **160/110 or more** = severe; hypotension = late sign of shock
Respiratory rate	**Unchanged** (12–20/min)	**Above 20/min** is abnormal: PE, pneumonia, pulmonary oedema, sepsis, DKA
SpO_2	95% or more	**Below 95%** is never physiological
Temperature	Up to about 0.5°C higher early on	**38.0°C or more** = fever
Shock index (HR ÷ SBP)	0.7–0.9	**0.9 or more** worrying; **1.4 or more** critical

3. **Positive hCG + pain or bleeding + tachycardia or shock index 0.9 or more** → treat as a **ruptured ectopic** until proven otherwise: call for help, two large-bore cannulae, group and cross-match, urgent scan or straight to theatre if unstable.
4. **From 20 weeks, never leave her supine**: left lateral tilt (15–30°) or manual uterine displacement before you examine, resuscitate or transfer.

Step 1: Focused history

(A) Confirming and dating a pregnancy

Ask	Why: the answer that changes the plan
First day of LNMP; was it a normal period (flow, duration)?	Naegele's EDD. A lighter "period" may have been implantation or threatened miscarriage bleeding, so dates are uncertain → early scan
Cycle length and regularity	Naegele assumes 28 days. Long cycles: add (cycle length − 28) days. Irregular → scan dating
Contraception: type (DMPA, implant, pill, IUD) and when stopped	Amenorrhoea may be the method, not a pregnancy: Depo-Provera can cause amenorrhoea without pregnancy; conceiving with an IUD in place raises ectopic risk; stopping the pill causes irregular cycles
Breastfeeding?	Lactational amenorrhoea can hide a new pregnancy
Symptoms: nausea, breast tenderness, frequency, quickening	Presumptive support. **Severe vomiting** suggests twins, **molar pregnancy** or hyperemesis → early scan, hCG, U&E

Ask	Why: the answer that changes the plan
Pain, bleeding, shoulder-tip pain, fainting	**Ectopic** or **miscarriage** must be excluded early → urgent scan, not routine booking
Home pregnancy test results	Timing and repeat testing
Previous pregnancies, ectopics, PID, tubal surgery, infertility	Ectopic risk → locate the pregnancy by TVS
Medical conditions and drugs	Teratogens (valproate, warfarin, ACE inhibitors, methotrexate, isotretinoin) → switch now; folic acid **5 mg** if on antiepileptics or with a previous NTD

(B) Is this symptom physiological? For every symptom ask: **onset** (gradual = more likely physiological; sudden = pathology), **severity** (does it stop her from normal activity?), **associated features** (the "red-flag companions" below), and **trend** (getting worse week by week?).

Symptom	Reassuring (physiological) companions	Red-flag companions (pathology)
Breathlessness	Gradual, on talking or mild exertion, no cough	Sudden, at rest, orthopnoea, PND, chest pain, haemoptysis, palpitations
Swelling	Both ankles, evening, settles overnight	Face or hands, headache, visual symptoms (pre-eclampsia); one leg (DVT)
Palpitations	Brief, no syncope	Syncope, chest pain, known RHD, sustained HR above 100
Vomiting	First trimester, eating and drinking something	Dehydration, weight loss, after 16 weeks (look for malaria, UTI, hepatitis, molar pregnancy)
Itching	Localised, with rash, stretch-mark area	**Palms and soles, no rash** (intrahepatic cholestasis)
Headache	Mild, early pregnancy, responds to paracetamol	After 20 weeks, severe, with visual disturbance or epigastric pain (pre-eclampsia)
Discharge	White, non-offensive, non-itchy	Itch, odour, colour, or watery (ruptured membranes)

Step 2: Focused examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for the pregnant woman.

Introduction, consent and preparation

- **Introduce yourself**, confirm her identity, explain the examination and obtain **consent**; offer a **chaperone** (always for the breast and pelvic examination).
- **Hand hygiene** before and after.
- **Position and exposure**: semi-recumbent; **from 20 weeks never flat on her back** (left lateral tilt 15–30°); expose only the part being examined.

General inspection and vital signs

- Well or unwell, pallor, jaundice, **oedema (where?)**, hydration, weight and height (BMI; baseline for weight gain).
- **Vital signs read against pregnancy norms** (Step 0 table): pulse, BP (baseline for later comparison), RR, SpO_2, temperature, shock index if bleeding.

Hands, face and neck

- **Hands**: pulse rate and character; palmar erythema (oestrogen, can be normal); capillary refill.

- **Face**: conjunctival pallor; chloasma; gum hyperaemia (pregnancy gingivitis, epulis).
- **Neck**: JVP; **thyroid** (modest symmetrical enlargement can be normal; a nodule is not).

Breasts (with consent and a chaperone)

Enlargement, **Montgomery's tubercles**, secondary areola, pigmented nipples, colostrum from about 16 weeks.

Cardiorespiratory

A **soft ejection systolic flow murmur** is found in most pregnant women and is normal. **A diastolic murmur, a loud (grade 3 or more) murmur, a displaced heaving apex, basal crepitations or an irregular pulse are never physiological.** Chest: clear, normal RR.

Abdomen

1. **Inspection**: linea nigra, striae, scars, size of the abdomen.
2. **Fundal height** against the landmarks (**12 weeks** just above symphysis, **16** halfway to umbilicus, **20–22** at umbilicus, **36** at xiphisternum); the fundus is palpable from 12 weeks.
3. **Fetal parts** (from about 20 weeks): lie and presentation later in pregnancy.
4. **Auscultation**: fetal heart with **Doppler from 10–12 weeks**, **Pinard from about 20 weeks**.

Pelvic examination (only when indicated)

- **Speculum first**: bleeding source, os open or closed, bluish cervix (Chadwick's), discharge.
- **Bimanual**: uterine size (a hen's egg is about 6 weeks, an orange about 8 weeks, a grapefruit about 12 weeks), softening (Hegar's/Goodell's), **adnexal tenderness or mass, cervical excitation** (ectopic).

Legs and completing the examination

- **Legs**: calf size and tenderness on both sides (DVT is commoner on the **left**); ankle oedema.
- Cover her, **thank her**, **hand hygiene**, and **summarise** with an impression ("physiological" or "needs investigation").

Document for a normal early pregnancy: "Pulse 88, BP 108/64, RR 16, SpO_2 98%, no pallor, soft ejection systolic murmur only, uterus about 8 weeks' size, soft, non-tender, no adnexal mass, os closed."

Step 3: Bedside tests

Test	Normal in pregnancy	What an abnormal result means
Urine hCG (detects about 20–25 IU/L)	Positive from about the missed period	Negative with strong suspicion → repeat in 1 week or serum β-hCG; very high hCG can cause a false negative (hook effect) in a mole
Urine dipstick	Trace glucose or protein can be normal	**Protein 1+ or more with BP 140/90** → pre-eclampsia workup; **nitrites/leucocytes** → UTI; **ketones** 2+ or more with vomiting → dehydration/hyperemesis
HemoCue Hb	**11 g/dL or more** (10.5 or more acceptable in the 2nd trimester)	Below 11 anaemia; below 7 severe
Pulse oximetry	**95% or more**	Below 95% = pathology (PE, pneumonia, pulmonary oedema)
RBS	Fasting tends lower in early pregnancy	High → formal FBS/OGTT (glycosuria alone is not enough)
Doppler / Pinard	FHR **110–160 bpm**	Not heard when expected → ultrasound
Bedside clotting test (2 mL in a plain glass tube)	Clot by about 7 minutes (pregnancy is hypercoagulable)	**No clot by 7 minutes** or a clot that breaks up → coagulopathy/DIC

Test	Normal in pregnancy	What an abnormal result means
Point-of-care ultrasound (if trained)	Intrauterine sac, fetal heart	Empty uterus + free fluid in the pouch of Douglas/Morison's pouch + positive hCG → ruptured ectopic

Step 4: Laboratory & imaging

Test	When	Expected (normal pregnancy) / abnormal finding	How it changes management (and Ugandan availability)
Serum quantitative β-hCG	Pregnancy of unknown location, suspected ectopic or mole, follow-up after miscarriage/GTD	Detectable 8–11 days after conception; **rises 63% or more in 48 h** in a viable early IUP; peaks at 8–10 weeks. **Above the discriminatory zone (about 1,500–3,500 IU/L) with an empty uterus on TVS** = ectopic or complete miscarriage	Decides between expectant follow-up, methotrexate, or surgery. Regional/national referral and private labs
Transvaginal ultrasound	Early pregnancy with pain or bleeding; dating	Sac about 4.5–5 weeks, yolk sac about 5.5, **cardiac activity about 6 weeks**; CRL dates to ±5–7 days at 7–14 weeks	Location, viability, number, dates. Hospitals and some HC IVs (transabdominal much more common; sees things about 1 week later)
Dating / anatomy ultrasound (transabdominal)	At least once **before 24 weeks**	Singleton or twins, placenta site, anomalies	Scan EDD replaces LNMP EDD if they differ by more than 5–7 days (1st trimester) or 10–14 days (2nd trimester). Most HC IVs and hospitals
FBC	Booking, anaemia, infection, pre-eclampsia	**Hb 11 g/dL or more**; **WCC up to about 15 × 10^9/L** (higher in labour and after steroids); platelets 150–400 (gestational thrombocytopenia usually above 100)	Platelets **below 100** → HELLP, ITP, DIC. Hb high for gestation in a hypertensive woman → poor plasma expansion (pre-eclampsia). HC IV and hospital
Creatinine and urea	Hypertension, sepsis, vomiting, before nephrotoxic drugs	**Creatinine 35–70 µmol/L**; above about 77–80 suspicious; **above 90 clearly abnormal**. Urea 2–4.5 mmol/L	A "normal" non-pregnant creatinine may mean renal impairment in pregnancy → pre-eclampsia with severe features, sepsis, dehydration. Hospital labs
LFTs	Vomiting after 16 weeks, jaundice, itching, pre-eclampsia	**ALP up 2–4× (placental), albumin low (28–37 g/L)** are normal. **ALT/AST above about 35 IU/L are abnormal**	Raised transaminases → HELLP, AFLP, cholestasis (with bile acids), viral hepatitis. Hospital labs
Fibrinogen, PT/APTT	PPH, abruption, IUFD, HELLP	**Fibrinogen 4–6 g/L**; PT/APTT normal or slightly short	**Fibrinogen below 2 g/L in PPH = severe coagulopathy** → blood products early. Regional/national referral only; use the bedside clotting test elsewhere
ABG	Breathlessness, asthma, sepsis, DKA	**pH about 7.42–7.44, $PaCO_2$ about 4.0 kPa (30 mmHg), HCO_3^- 18–22 mmol/L**	$PaCO_2$ of 5.3 kPa or more = respiratory fatigue → HDU/ICU. Mainly national referral/ICU
TFTs	Hyperemesis, goitre, palpitations, known thyroid disease	TSH lower in 1st trimester (about 0.1–2.5 mU/L); **free** T4 normal (total T4 up with TBG)	Low TSH with high hCG in hyperemesis is usually **transient gestational thyrotoxicosis**: do not start antithyroid drugs without a senior. Regional/national referral

Test	When	Expected (normal pregnancy) / abnormal finding	How it changes management (and Ugandan availability)
75 g OGTT	24–28 weeks if risk factors	Fasting below 5.1 mmol/L normal; GDM if fasting 5.1 or more, 1 h 10.0 or more, or 2 h 8.5 or more	Insulin resistance of late pregnancy unmasks GDM. Regional/national referral
D-dimer, ESR, CRP	Rarely useful	**All rise in normal pregnancy**	A positive D-dimer does not diagnose VTE: go straight to **compression Doppler ultrasound** (and CTPA/V/Q where available) if clinical suspicion
ECG / echocardiogram	Palpitations, murmur with symptoms, breathlessness at rest	ECG may show **left axis deviation, T inversion in III, ectopics** normally	Echo for any diastolic murmur (mitral stenosis) or suspected peripartum cardiomyopathy. Echo at regional/national referral (e.g. Uganda Heart Institute, Mulago)

For all of these tests (urine and serum hCG, TVS and transabdominal ultrasound, Doppler and Pinard, FBC, renal and liver tests, clotting, ABG, TFTs and OGTT), two rules apply: **urine hCG first** in any woman of reproductive age, and **interpret every result against pregnancy-specific ranges**.

Step 5: Putting it together

Model one-line summary (physiology vs pathology):

> "Mrs N., 27-year-old G2P1 at **32 weeks**, with **2 weeks of gradual breathlessness on exertion**. No orthopnoea, chest pain or cough. Pulse 92, BP 112/70, **RR 16**, **SpO_2 98%**, soft ejection systolic murmur only, chest clear, ankle oedema only, Hb 11.4 g/dL. **Impression: physiological breathlessness of pregnancy**; no evidence of cardiac or respiratory disease."

Problem list (for the same patient): 1) breathlessness, likely physiological; 2) ankle oedema with normal BP and urine; 3) routine ANC due (IPTp, Hb recheck). **Working diagnosis:** normal singleton pregnancy at 32 weeks with physiological symptoms. **Plan:** reassure, safety-net (come back at once with breathlessness at rest, chest pain, swelling of the face or one leg, or headache).

Worked example 1: reading an ABG in a pregnant woman

	Patient X (well, 30 weeks)	Patient Y (asthma attack, 30 weeks)
pH	7.44	7.38
$PaCO_2$	4.0 kPa (30 mmHg)	**5.4 kPa (41 mmHg)**
HCO_3^-	20 mmol/L	22 mmol/L
Interpretation	**Normal compensated respiratory alkalosis of pregnancy**	$PaCO_2$ looks "normal" for a non-pregnant adult but is **high for pregnancy**: she is **tiring**. Escalate to HDU/ICU and call the anaesthetist

Worked example 2: "normal" results that are abnormal in pregnancy

A 34-week woman with BP 150/96 mmHg has: **creatinine 92 µmol/L**, **ALT 68 IU/L**, **ALP 310 IU/L**, **platelets 96 × 10^9/L**, Hb 13.4 g/dL, albumin 29 g/L.

- Creatinine 92 is within many non-pregnant ranges but **clearly abnormal in pregnancy** (normal 35–70) → renal involvement.
- ALT 68 is **abnormal** (transaminases do not rise in normal pregnancy).
- ALP 310 is **expected** (placental ALP rises 2–4×): do not chase it.

- Platelets 96 are **below 100** → not gestational thrombocytopenia in this context.
- Hb 13.4 at 34 weeks is **high for gestation** → poor plasma expansion (haemoconcentration).
- Albumin 29 g/L is within the normal pregnancy range.
- **Conclusion:** pre-eclampsia with severe features (renal, hepatic and haematological involvement), possible evolving HELLP → admit, $MgSO_4$, control BP, plan delivery (see the pre-eclampsia topic).

> **CLINICAL PEARL**
> When you present any pregnant woman's results, say the pregnancy-normal range out loud: "Her creatinine is 92, which is **high for pregnancy** where we expect 35 to 70." It shows the consultant you know the physiology and stops the team being falsely reassured.

Model summary line (diagnosing pregnancy)

> "Mrs K is a **26-year-old G1P0** who presents with **7 weeks of amenorrhoea**, nausea and breast tenderness. Her urine hCG is positive and a transvaginal scan shows a **single intrauterine gestational sac with a fetal pole of CRL 11 mm and cardiac activity**, consistent with **7 weeks + 2 days**, matching her LNMP dates. **Diagnosis: a viable singleton intrauterine pregnancy at 7 weeks.** Plan: book ANC, start iron–folate, booking bloods, and advice for nausea."

5. Clinical Acumen

Thinking like a clinician

- **"Every woman of reproductive age is pregnant until proven otherwise."** Before an X-ray, prescription, laparotomy or diagnosis of "PID" or "appendicitis", do a urine hCG.
- **Interpreting hCG with ultrasound:**
  - hCG above the discriminatory zone (about 1,500–3,500 IU/L) with **no intrauterine sac on TVS** suggests ectopic or complete miscarriage.
  - hCG below the zone with no sac means a **pregnancy of unknown location**: repeat hCG in 48 hours. A rise of 63% or more suggests a viable IUP; a fall of 50% or more suggests failing pregnancy; a suboptimal rise suggests ectopic.
  - Very high hCG suggests molar pregnancy, twins, or wrong dates.
- **Negative test but still suspect pregnancy?** Too early (repeat in 1 week), dilute urine (use the first morning sample), or the **hook effect** (extremely high hCG in a mole can give a false negative; dilute the sample).
- **Positive test but not pregnant?** Recent pregnancy or miscarriage (hCG can persist about 4–6 weeks), hCG injection (infertility treatment), gestational trophoblastic disease, rare tumours (germ cell).
- **Physiology vs pathology:** before attributing anything to "normal pregnancy", ask "**Is this within the expected range of change, and is the trend right?**"
  - Breathlessness: physiological if gradual, at rest or on talking, with a normal RR, SpO_2 and exam. Pathological if sudden, with tachypnoea, tachycardia, chest pain, cough or haemoptysis.
  - Oedema: physiological if bilateral ankle oedema, worse in the evening, with normal BP. Pathological if facial or hand oedema with hypertension, or unilateral.
  - Murmur: physiological if soft systolic ejection. Pathological if diastolic, loud (grade 3 or more), or with symptoms.
  - Hb: 10.5–11 in the second trimester may be mild dilution, but **below 11 is still treated** as anaemia in Uganda (WHO cut-off), and below 10 is almost always true anaemia.

- **Haemorrhage physiology pitfall:** because of the expanded blood volume, a pregnant woman can lose **1,000–1,500 mL (up to about 30% of blood volume) before BP falls. Tachycardia and a narrowing pulse pressure come first**; hypotension is a late, pre-arrest sign. Use the **shock index** (HR ÷ systolic BP): **0.9 or more is worrying, 1.4 or more is critical**.

Red flags

RED FLAG

- **Positive pregnancy test + pain or bleeding + empty uterus on scan** means **ectopic until proven otherwise**. Shock, shoulder-tip pain or peritonism means a ruptured ectopic: resuscitate and take her to theatre.
- **Uterus much larger than dates + very high hCG + hyperemesis or early pre-eclampsia (before 20 weeks) + vaginal bleeding** means a **molar pregnancy**.
- **Breathlessness with RR above 20/min, SpO_2 below 95%, orthopnoea, a diastolic murmur, or tachycardia above 100 bpm** is **not** physiological: think cardiac disease (RHD, peripartum cardiomyopathy), PE, pneumonia or severe anaemia.
- **Unilateral leg swelling** (usually left) means DVT until proven otherwise.
- **Generalised or facial oedema with hypertension** means pre-eclampsia.
- **Pruritus of the palms and soles, no rash** means intrahepatic cholestasis (check bile acids and LFTs).

Amenorrhoea in a woman of reproductive age

Condition	Distinguishing features	Key investigation
Intrauterine pregnancy	Symptoms of pregnancy, enlarging soft uterus	Urine/serum hCG; ultrasound
Ectopic pregnancy	Amenorrhoea + pain + spotting; adnexal tenderness; uterus smaller than dates	hCG + TVS (empty uterus, adnexal mass, free fluid)
Molar pregnancy	Uterus larger than dates, hyperemesis, bleeding, early pre-eclampsia, theca-lutein cysts	Very high hCG; "snowstorm" ultrasound
Hormonal contraception (DMPA, implant)	History of injection; hCG negative	hCG
Lactational amenorrhoea	Breastfeeding under 6 months	hCG to exclude a new pregnancy
PCOS	Oligomenorrhoea, hirsutism, obesity, acne	hCG negative; ultrasound ovaries; hormones
Hyperprolactinaemia	Galactorrhoea, headaches, visual field loss	Prolactin, hCG negative
Premature ovarian insufficiency / menopause	Hot flushes, age	FSH raised
Pseudocyesis	Firm belief of pregnancy, abdominal distension, no fetus	hCG negative, ultrasound empty
Stress, weight loss, chronic illness (TB, HIV)	Context	hCG negative

Uterus larger than dates

Wrong dates, **twins**, **molar pregnancy**, **polyhydramnios**, **fibroids**, **macrosomia**, full bladder.

Common student mistakes

- Labelling abdominal pain as "PID" or "appendicitis" without a **urine hCG**.

- Reading a pregnant woman's creatinine, $PaCO_2$ or fibrinogen against **non-pregnant** ranges.
- Calling breathlessness "physiological" without counting the **RR** and checking **SpO_2**.
- Ordering a **D-dimer** to exclude VTE in pregnancy.
- Waiting for **hypotension** before calling a bleeding pregnant woman shocked.
- Examining or resuscitating a woman beyond 20 weeks **flat on her back**.

6. Common Problems and Their Management

This topic is about normal physiology, so "management" means **(1) confirming and dating the pregnancy, (2) managing common physiological symptoms safely, and (3) applying physiology to emergencies.**

1. Confirming and dating a pregnancy: step by step

1. **Take the menstrual history** and calculate the EDD by Naegele's rule (adjust for cycle length).
2. **Do a urine hCG test.** If negative but suspicion is high, repeat in 1 week or send serum β-hCG.
3. **If positive with pain or bleeding:** vital signs first, then urgent ultrasound. Manage a possible ectopic as an emergency (see the ectopic topic).
4. **If positive and well:** book ANC (ideally **before 12 weeks**). Start **folic acid 400 µg** (with iron per the Uganda ANC package), or **5 mg daily** if she has a previous NTD-affected baby, takes antiepileptics, has diabetes, sickle cell disease or BMI 30 or more.
5. **Dating ultrasound** (CRL at 7–14 weeks is best; WHO recommends one scan before 24 weeks). If the ultrasound and LNMP EDDs differ by more than 5–7 days in the first trimester (or more than 10–14 days in the second trimester), **use the scan date**.
6. **Record the EDD** on the ANC card and **never change it later** based on late scans or fundal height.

2. Managing common "physiological" complaints

Complaint	Physiology	Management
Nausea and vomiting	hCG, oestrogen	Small frequent dry meals, ginger, avoid triggers. If needed: **promethazine 25 mg orally 8-hourly** or **metoclopramide 10 mg orally 8-hourly**; pyridoxine (vitamin B6) 10–25 mg 8-hourly. Hyperemesis (dehydration, ketonuria, weight loss): admit for **IV normal saline with potassium**, **thiamine 100 mg** before any dextrose (Wernicke's), and IV/IM antiemetic
Heartburn	Progesterone relaxes the LOS; uterine pressure	Small meals, avoid lying flat after meals, raise the head of the bed. **Magnesium trisilicate** 10 mL 8-hourly or an aluminium hydroxide antacid (separate from iron by 2 hours)
Constipation	Progesterone slows transit; iron	Fluids, fruit and vegetables; lactulose 15 mL 12-hourly if needed
Haemorrhoids and varicose veins	Venous compression, progesterone	Avoid constipation; leg elevation; support stockings
Ankle oedema	Venous pressure, sodium retention, low albumin	Leg elevation, left lateral rest. **Always check BP and urine**
Backache	Lordosis, relaxin	Posture, flat shoes, paracetamol 1 g 6–8-hourly (max 4 g/day). **Avoid NSAIDs**, especially after 20 weeks

Complaint	Physiology	Management
Leg cramps	Uncertain	Stretching, hydration
Urinary frequency	Uterine pressure, increased GFR	Reassure, but **dipstick** to exclude UTI
Physiological breathlessness	Progesterone-driven hyperventilation	Reassure only after a normal RR, SpO_2, chest and heart exam
Vaginal discharge	Oestrogen-driven leucorrhoea	Reassure if white, non-itchy, non-offensive. Treat candidiasis with **clotrimazole 500 mg pessary single dose** (or 100 mg nightly for 7 nights; longer courses work better in pregnancy). Avoid oral fluconazole in pregnancy
Syncope / dizziness	Low SVR, aortocaval compression	Avoid lying supine and standing up quickly; **lie on the left side**

DRUG DOSING

Drugs in pregnancy that are generally safe: paracetamol, amoxicillin, ampicillin, cephalosporins, erythromycin/azithromycin, metronidazole (after the first trimester is traditional, but it can be used), nitrofurantoin (avoid at term), methyldopa, nifedipine, labetalol, hydralazine, insulin, heparin, TLD, artemether–lumefantrine (all trimesters per updated WHO guidance), quinine. **Avoid:** ACE inhibitors/ARBs, warfarin (first trimester and near term), tetracyclines/doxycycline, fluoroquinolones (if alternatives exist), valproate, isotretinoin, methotrexate, misoprostol (unless indicated), NSAIDs in the third trimester (premature closure of the ductus arteriosus, oligohydramnios), statins, and iodine-131.

3. Applying physiology in emergencies: the "pregnancy-adapted" ABC

1. **Position:** left lateral tilt of 15–30° or **manual left uterine displacement** from 20 weeks, to relieve aortocaval compression.
2. **Airway:** expect a difficult airway (oedema, full stomach), so do a **rapid-sequence induction** with a smaller endotracheal tube and give **antacid prophylaxis** (e.g. sodium citrate 30 mL, or ranitidine/omeprazole) before CS.
3. **Breathing: pre-oxygenate** well (reduced FRC means rapid desaturation). Aim for **SpO_2 of 95% or more**.
4. **Circulation:** remember the blood-volume cushion. **Treat tachycardia as haemorrhage until proven otherwise**, use two large-bore cannulae, and do not wait for hypotension.
5. **Coagulation:** in PPH or abruption, watch fibrinogen (it should be high). **Below 2 g/L predicts severe PPH**, so give tranexamic acid and blood products early.
6. **Thromboprophylaxis:** after CS or in immobile, high-risk women give **enoxaparin 40 mg SC daily** (weight-adjusted) or unfractionated heparin 5,000 IU SC 12-hourly where available. Mobilise early and keep her hydrated.
7. **Perimortem CS:** in maternal cardiac arrest with a uterus at or above the umbilicus, start CPR with left uterine displacement. If there is no return of circulation, **deliver by 4–5 minutes** to improve maternal resuscitation.

UGANDA CONTEXT

In many Ugandan HC IVs, **ABG, fibrinogen and D-dimer are unavailable**. Rely on the bedside: HR, RR, SpO_2, shock index, urine output, capillary refill, conjunctival pallor, and the **bedside clotting test** (WHO: 2 mL of venous blood in a small plain glass tube, tilted every minute; **no clot by 7 minutes**, or a soft clot that breaks up easily, suggests coagulopathy/DIC; some units use the 20-minute whole-blood clotting test instead). **D-dimer rises normally in pregnancy**, so a positive result is unhelpful.

4. Complications: when the adaptation fails

What happens when the physiological adaptation **fails or is exceeded**:

Normal change	When it goes wrong
Plasma volume expansion	Poor expansion goes with **pre-eclampsia** and FGR. Excess demand on a diseased heart causes **pulmonary oedema** in mitral stenosis or peripartum cardiomyopathy
Increased cardiac output	Decompensation of **rheumatic heart disease** at 28–32 weeks, in labour and immediately postpartum
Hypercoagulability	**DVT/PE** (a leading direct cause of maternal death worldwide), **cerebral venous thrombosis**
Insulin resistance	**Gestational diabetes**, macrosomia, shoulder dystocia
Ureteric dilatation and stasis	**Pyelonephritis**, preterm labour, sepsis
Reduced FRC and gastric emptying	**Hypoxia** during fits or anaesthesia; **aspiration pneumonitis** (Mendelson's syndrome)
Aortocaval compression	**Supine hypotension**, fetal distress; failed resuscitation
Uterine blood flow of about 700 mL/min	**Massive PPH** within minutes
Increased thyroid demand	Unmasked hypothyroidism or goitre
Pituitary enlargement	**Sheehan's syndrome** after PPH-induced hypotension

7. Guidelines, Policy and Law in Uganda

UGANDA CONTEXT

This topic follows the **Uganda Clinical Guidelines (UCG) 2023**, the Uganda ANC package and the **WHO 2016 recommendations on antenatal care for a positive pregnancy experience** (at least 8 contacts, the first before 12 weeks; **one ultrasound before 24 weeks**; daily iron and folic acid).

- **Booking**: ANC ideally **before 12 weeks**; every woman of reproductive age with pain, bleeding or collapse gets a **urine hCG** first.
- **Folic acid and iron**: folic acid **400 µg** daily with iron per the Uganda ANC package; **5 mg** for a previous NTD-affected baby, antiepileptics, diabetes, sickle cell disease or BMI 30 or more.
- **Dating**: the scan EDD replaces the LNMP EDD if they differ by more than 5–7 days (first trimester) or 10–14 days (second trimester); the EDD is then **fixed** on the ANC card.
- **GDM screening**: 75 g OGTT at **24–28 weeks** if risk factors; GDM if fasting 5.1 mmol/L or more, 1 h 10.0 or more, or 2 h 8.5 or more (WHO 2013 criteria).
- **Drugs**: the safe and avoid lists in section 6; artemether–lumefantrine is acceptable in all trimesters per updated WHO malaria guidance.

- **Emergencies**: left uterine displacement, the shock index and the bedside clotting test are the tools available at every level (section 6); ABG, fibrinogen and D-dimer are mostly unavailable below regional referral level.

> UGANDA CONTEXT
> **Where UCG and WHO/international guidance differ**: the anaemia cut-off. UCG and Ugandan practice treat **Hb below 11 g/dL** as anaemia throughout pregnancy, whereas the **WHO 2024 haemoglobin guideline** uses **below 11.0 g/dL in the first and third trimesters and below 10.5 g/dL in the second trimester** Verify *confirm the trimester-specific WHO 2024 cut-offs and whether UCG 2023 has adopted them*. Quote "below 11" in Ugandan exams and know the WHO nuance.

8. Key Points

High-yield points

- **Pregnant until proven otherwise**: urine hCG in every woman of reproductive age with pain, bleeding, collapse, or before a drug, X-ray or operation.
- **Positive signs only come from the fetus**: heart heard, parts or movements felt by the examiner, fetus seen on ultrasound.
- **hCG above about 1,500–3,500 IU/L with an empty uterus on TVS** = ectopic until proven otherwise.
- **The 50% club**: plasma volume, cardiac output, minute ventilation and GFR all rise by roughly half.
- **Normal ranges change**: creatinine 35–70 µmol/L, $PaCO_2$ about 4 kPa, fibrinogen 4–6 g/L, ALP up 2–4 times.
- **Never physiological**: RR above 20, SpO_2 below 95%, resting HR above 100, a diastolic murmur, raised ALT/AST, unilateral leg swelling.
- **Tilt her left from 20 weeks.**
- **She keeps her BP until about 30% blood loss**: watch the pulse and the shock index.
- **Hypercoagulable, highest postpartum**: VTE risk about 5-fold.
- **D-dimer, ESR and WCC rise in normal pregnancy**: do not be misled.

Numbers & doses

Item	Value
Urine hCG positive	About the day of the missed period (strip detects about 20–25 IU/L)
Serum β-hCG	Detectable 8–11 days after conception; rises **63% or more in 48 h** in a viable IUP; peaks **8–10 weeks**
Discriminatory zone (TVS)	hCG about **1,500–3,500 IU/L**
TVS milestones	Sac **4.5–5 weeks**; yolk sac **5.5 weeks**; cardiac activity **6 weeks**
Fetal heart	Doppler **10–12 weeks**; Pinard about **20 weeks**
Quickening	**18–20 weeks** (primigravida); **16–18 weeks** (multigravida)
Fundal landmarks	12 weeks symphysis; 16 halfway; **20–22 umbilicus**; **36 xiphisternum**
Naegele's rule	LNMP + 7 days − 3 months + 1 year (+ 280 days)
Re-date by scan if LNMP differs by	More than **5–7 days** (1st trimester) or **10–14 days** (2nd trimester)
Plasma volume / red cell mass	Up **40–50%** / up **20–30%**
Cardiac output	Up **30–50%**; up to **60–80%** in labour and just after delivery

Item	Value
Heart rate	Up **10–20 bpm**; above 100 at rest is abnormal
Hb (anaemia cut-off)	Below **11 g/dL**
WCC	Up to about **15 × 10^9/L**; up to about 25 in labour
Fibrinogen	**4–6 g/L**; below 2 g/L in PPH = severe coagulopathy
GFR	Up about **50%**; creatinine **35–70 µmol/L**
ABG	pH about **7.44**, $PaCO_2$ about **4.0 kPa (30 mmHg)**, HCO_3^- **18–22 mmol/L**
FRC / O_2 consumption	Down about **20%** / up about **20%**
Uterine blood flow at term	About **500–750 mL/min**
Weight gain (normal BMI)	**11–16 kg**
Aortocaval compression	From about **20 weeks**: tilt **15–30° left**
Perimortem CS	Deliver within **4–5 minutes** of arrest if no ROSC (uterus at or above umbilicus)

Exam traps

- **"Hb 10.5 g/dL at 28 weeks is just dilution, so no treatment."** → Physiological haemodilution is real, but in Uganda (WHO) **below 11 g/dL is anaemia** and is treated and investigated.
- **"$PaCO_2$ 5.3 kPa is normal."** → Normal for a non-pregnant adult, but **high for pregnancy** (normal about 4 kPa): in an asthmatic it signals **respiratory fatigue**.
- **"Creatinine 90 µmol/L is normal."** → In pregnancy creatinine should be **35–70**; 90 is **clearly abnormal** (think pre-eclampsia, sepsis, dehydration).
- **"Raised ALP means liver disease."** → ALP rises **2–4×** from the placenta. It is **ALT/AST** that should never rise.
- **"Breathlessness in pregnancy is physiological."** → Only with a **normal RR, SpO_2 and examination**. A raised RR or low SpO_2 is pathology (PE, pneumonia, pulmonary oedema, mitral stenosis).
- **"Positive D-dimer, so she has a PE."** → D-dimer rises in normal pregnancy. Use **compression Doppler / imaging**.
- **"Glycosuria means diabetes."** → The renal threshold falls in pregnancy; diagnose GDM only by **blood glucose (OGTT)**.
- **"BP is normal, so she is not bleeding much."** → Pregnant women keep their BP until about 30% blood loss: **tachycardia and a rising shock index come first**.

Questions seniors ask

Viva questions

1. **What are the positive signs of pregnancy?** Fetal heart heard, fetal movements or parts felt by the examiner, and the fetus visualised on ultrasound.
2. **Why is a pregnant woman's Hb lower?** Plasma volume rises about 50% but red cell mass only about 25%, so haemodilution occurs. It lowers viscosity and improves placental perfusion.
3. **Why does a pregnant woman desaturate quickly?** FRC is down about 20% (elevated diaphragm) and O_2 consumption is up about 20%, so the oxygen reserve is used up quickly during apnoea.
4. **Why are pregnant women prone to pyelonephritis?** Progesterone-induced ureteric dilatation and pressure at the pelvic brim cause urinary stasis, and glycosuria adds to it. Asymptomatic bacteriuria therefore ascends.
5. **What ABG do you expect in normal pregnancy?** Compensated respiratory alkalosis: pH about 7.44, $PaCO_2$ about 4 kPa, HCO_3^- about 18–22 mmol/L.

6. **Why does BP fall in mid-pregnancy?** Progesterone and nitric oxide cause vasodilation, and the low-resistance placental circulation lowers SVR, with the nadir at about 20–24 weeks.
7. **When is a heart murmur in pregnancy abnormal?** When it is diastolic, grade 3 or more, radiates widely, or comes with symptoms or signs (orthopnoea, arrhythmia, cyanosis).
8. **How long can a urine pregnancy test stay positive after delivery or miscarriage?** Up to about 4–6 weeks as hCG clears.

On the ward round

Name the positive signs of pregnancy. Fetal heart heard, fetal parts or movements felt by the examiner, and the fetus seen on ultrasound. Everything else (including hCG) is presumptive or probable.

The hCG is 2,800 IU/L and the TVS shows an empty uterus. What do you think? hCG is above the discriminatory zone, so an intrauterine sac should usually be visible: **ectopic pregnancy** (or a complete miscarriage) until proven otherwise. Check for an adnexal mass and free fluid; if she is stable, repeat hCG in 48 hours with a senior plan; if unstable, theatre.

Why does a pregnant woman desaturate so quickly? FRC falls about 20% and O_2 consumption rises about 20%, so her oxygen reserve is small and used up fast: pre-oxygenate and protect the airway early.

Which murmur in pregnancy worries you? Any **diastolic** murmur, a murmur of grade 3 or more, or any murmur with symptoms. In Uganda think **rheumatic mitral stenosis**: get an echo.

When does a woman with heart disease decompensate, and why? At **28–32 weeks** (cardiac output near peak), **in labour**, and **immediately after delivery** (autotransfusion of about 500 mL from the contracting uterus).

Why must she lie tilted to the left? From about 20 weeks the supine uterus compresses the IVC and aorta, cutting venous return and cardiac output by up to 25–30%: supine hypotension and fetal distress.

Why are pregnant women prone to VTE, and when is the risk highest? Virchow's triad: stasis (venous compression, progesterone), hypercoagulability (fibrinogen, VII, VIII, X up; protein S down) and vessel injury at delivery. Risk is about 5-fold, **highest postpartum**, especially after CS.

Mnemonics

Things that go UP: "the 50% club"

Roughly **half again**: **plasma volume** (40–50%), **cardiac output** (30–50%), **minute ventilation** (40–50%), **GFR and renal plasma flow** (50% or more). Smaller rises: red cell mass (20–30%), heart rate (10–20 bpm), tidal volume, O_2 consumption (20%), WCC, ESR, fibrinogen and clotting factors VII, VIII and X, ALP, cortisol, TBG, prolactin.

Things that go DOWN: "Down with SPICE-FR"

- **S**VR (and so mid-pregnancy BP)
- **P**aCO_2 (and so HCO_3^-)
- **I**nsulin sensitivity (second half)
- **C**reatinine and urea
- **E**xpiratory reserve volume
- **F**RC
- **R**ed cell concentration (Hb), plus albumin and protein S

Signs of early pregnancy by the "letters of the alphabet": Chadwick (Colour, bluish), **G**oodell (Gentle cervix, soft), **H**egar (Hollow feel of the isthmus, soft). **C, G, H** come in alphabetical order, roughly matching the order in which you would examine: look (C), then feel the cervix (G), then bimanual (H).

Fetal heart detection "10–20" rule: Doppler at about **10** weeks, Pinard at about **20** weeks.

Fundal height "12–16–20–36": symphysis at 12 weeks, halfway at 16, umbilicus at 20 (to 22), xiphisternum at 36.

OSCE tips

- When asked to "confirm pregnancy", state: history (LNMP, symptoms), a **urine hCG**, and an ultrasound (location, viability, number, dating).
- Calculate an EDD aloud: LNMP **10 March 2025**: add 7 days (17 March 2025), subtract 3 months (17 December 2024), add 1 year, which gives an **EDD of 17 December 2025**.
- Explain physiology in patient-friendly language when counselling ("Your body makes extra blood for the baby; the water part increases more, so your blood level looks lower. That is why we give iron").
- **Physiology viva**: give each change as "**what changes, why, and what it means clinically**" (for example: FRC falls 20% because the diaphragm rises, so she desaturates quickly during a fit).

> **EXAM FAVOURITE**
> A favourite: "**A 30-week pregnant woman has a $PaCO_2$ of 5.5 kPa during an asthma attack. Comment.**" Answer: this is **high for pregnancy** (normal is about 4 kPa) and means **respiratory fatigue and impending failure**. Escalate to HDU/ICU.

9. References

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Presumptive, probable and positive signs of pregnancy	**Presumptive:** amenorrhoea, nausea, breast changes, frequency, quickening, skin changes. **Probable:** uterine enlargement, Hegar's, Goodell's, Chadwick's, ballottement, Braxton Hicks, **positive hCG**. **Positive: fetal heart heard**, **fetal parts or movements felt by the examiner**, **fetus seen on ultrasound**.
2	Hegar's, Goodell's and Chadwick's signs	**Hegar's:** softening of the uterine **isthmus** on bimanual exam (6–10 weeks). **Goodell's:** softening of the **cervix** (about 6–8 weeks). **Chadwick's: bluish discolouration** of the vagina and cervix from congestion (6–8 weeks).
3	Naegele's rule	EDD = **LNMP + 7 days – 3 months + 1 year** (= LNMP + 280 days). Assumes a regular 28-day cycle; for a longer cycle, add (cycle length – 28) days.
4	When does a pregnancy test become positive?	**Serum β-hCG:** 8–11 days after conception. **Urine hCG** (detects about 20–25 IU/L): around the **day of the missed period**.
5	hCG kinetics in early pregnancy	Doubles about every **48 hours** (a rise of at least about 63% in 48 hours suggests a viable IUP), **peaks at 8–10 weeks** (about 100,000 IU/L), then falls to a plateau. Very high in **molar pregnancy and twins**.
6	TVS milestones in early pregnancy	Gestational sac **about 4.5–5 weeks**; yolk sac **about 5.5 weeks**; fetal pole with **cardiac activity about 6 weeks**. Transabdominal scans lag about 1 week behind.
7	When can the fetal heart be heard with a Doppler and with a Pinard?	**Doppler:** about **10–12 weeks**. **Pinard stethoscope:** about **20 weeks** (18–24).
8	When does quickening occur?	**18–20 weeks** in a primigravida; **16–18 weeks** in a multigravida.
9	Plasma volume vs red cell mass in pregnancy	Plasma volume **up 40–50%**; red cell mass **up 20–30%**, giving **physiological haemodilution** (lowest Hb at about 28–34 weeks). WHO: **Hb below 11 g/dL = anaemia** in pregnancy.
10	Cardiac output in pregnancy, labour and postpartum	Up **30–50%** by the second trimester (SV first, then HR +10–20 bpm). A further rise in labour and **immediately after delivery (up to 60–80%)** from autotransfusion, the danger period for heart disease.
11	What happens to BP during normal pregnancy?	It falls in the first half (SVR down), with a **nadir at about 20–24 weeks** (diastolic falls most), and returns to baseline by term. Hypertension before 20 weeks suggests **chronic hypertension** or a **molar pregnancy**.
12	Normal ABG in pregnancy	**Compensated respiratory alkalosis**: pH about 7.42–7.44, **$PaCO_2$ about 4.0 kPa (30 mmHg)**, **HCO_3^- about 18–22 mmol/L**, PaO_2 slightly raised.
13	Respiratory changes: which go up, which stay the same, which go down?	**Up:** tidal volume (30–50%), minute ventilation (40–50%), O_2 consumption (20%). **Same:** respiratory rate, vital capacity. **Down:** FRC (about 20%), ERV, residual volume, $PaCO_2$.
14	Renal changes in pregnancy	RPF up 50–80%, **GFR up about 50%**, so **creatinine and urea fall** (creatinine above about 77 µmol/L is suspicious). **Glycosuria** is common. **Hydroureter/hydronephrosis, right more than left**, cause stasis and pyelonephritis risk.

#	Question	Answer
15	Coagulation changes in pregnancy	Up: **fibrinogen** (4–6 g/L), factors **VII, VIII, IX, X**, vWF, PAI-1 and PAI-2. Down: **protein S**, fibrinolysis. Result: **hypercoagulable**, VTE risk about 5-fold, highest postpartum.
16	Aortocaval compression: when, what, and what do you do?	From **about 20 weeks** the supine uterus compresses the IVC, reducing venous return and CO (up to 25–30%), causing supine hypotension and fetal distress. **Left lateral tilt 15° or manual left uterine displacement.**
17	Which LFT normally rises in pregnancy, and which never should?	**ALP rises 2–4 times** (placental isoenzyme). **Albumin falls. ALT/AST should not rise**; if raised, think HELLP, AFLP, obstetric cholestasis or hepatitis.
18	Thyroid function in pregnancy	**TBG up** means total T4/T3 are up but **free T4 is normal**. **TSH falls in the first trimester** (hCG has TSH-like activity). Use trimester-specific ranges.
19	Uterine size and blood flow at term	About **1,000–1,100 g** (from 50–70 g), capacity about **5 L**. Blood flow **about 500–750 mL/min** (10–15% of CO), which is why PPH can exsanguinate within minutes.
20	Why does a pregnant woman stay normotensive despite heavy bleeding?	Expanded blood volume (about +1.5 L) and sympathetic compensation: she may lose **1–1.5 L (up to about 30%)** before hypotension. **Tachycardia and a rising shock index (HR/SBP of 0.9 or more)** come first; hypotension is a late, pre-arrest sign.
21	Which vital signs are NEVER physiological in pregnancy?	**RR above 20/min**, **SpO_2 below 95%**, **resting HR above 100/min**, BP **140/90 or more**, temperature **38.0°C or more**, shock index **0.9 or more**. Each needs a cause found (PE, pneumonia, sepsis, bleeding, pre-eclampsia, anaemia, heart disease).
22	Normal creatinine in pregnancy, and what does 90 µmol/L mean?	Normal about **35–70 µmol/L** (GFR up about 50%). Above about 77–80 is suspicious; **90 µmol/L is clearly abnormal** in pregnancy (pre-eclampsia with severe features, sepsis, dehydration), even though it looks normal on a non-pregnant range.
23	hCG 2,800 IU/L with an empty uterus on TVS: interpretation?	hCG is **above the discriminatory zone (about 1,500–3,500 IU/L)**, so an intrauterine sac would usually be seen. Think **ectopic pregnancy** (or complete miscarriage). Look for an adnexal mass and free fluid; resuscitate and go to theatre if unstable.
24	Which LFT abnormalities are expected in pregnancy, and which are not?	**Expected:** ALP up **2–4×** (placental) and albumin down (about 28–37 g/L). **Never normal:** raised **ALT/AST** (above about 35 IU/L) or bilirubin → HELLP, AFLP, cholestasis, hepatitis.
25	Why is D-dimer unhelpful for suspected VTE in pregnancy, and what do you do instead?	D-dimer (like ESR and CRP) **rises in normal pregnancy**, so a positive result means little. If clinical suspicion: **compression Doppler ultrasound** of the leg (and CTPA or V/Q for PE where available). Start treatment-dose anticoagulation while waiting if suspicion is high, per senior/protocol.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 24-year-old woman presents to a HC IV with 6 weeks of amenorrhoea, mild left iliac fossa pain and spotting. BP 112/70 mmHg, pulse 88 bpm. Urine hCG is positive. What is the MOST appropriate next step?

A Reassure her and book ANC in 4 weeks
B Pelvic ultrasound (preferably transvaginal) to locate the pregnancy
C Give misoprostol for an incomplete miscarriage
D Treat for PID with ceftriaxone and doxycycline
E Laparotomy immediately

2. A 27-year-old woman's LNMP was 20 January 2025. Her cycles are regular, every 28 days. What is her expected date of delivery?

A 13 October 2025
B 20 October 2025
C 27 September 2025
D 27 October 2025
E 3 November 2025

3. A 30-year-old G2P1 at 28 weeks complains of feeling breathless when talking and climbing stairs for 3 weeks. RR 16/min, SpO_2 98% on air, HR 88 bpm, a soft ejection systolic murmur at the left sternal edge, clear chest, no oedema. What is the MOST likely explanation?

A Physiological dyspnoea of pregnancy from progesterone-driven hyperventilation
B Mitral stenosis
C Pulmonary embolism
D Peripartum cardiomyopathy
E Severe anaemia

4. A 32-week pregnant woman with asthma is in the emergency unit, wheezy and tired. ABG on oxygen: pH 7.38, $PaCO_2$ 5.4 kPa (41 mmHg), PaO_2 11 kPa. How should you interpret the $PaCO_2$?

A Normal; no concern
B Low, as expected from hyperventilation
C Raised for pregnancy, suggesting respiratory fatigue and impending failure
D Reflects metabolic alkalosis
E Normal for the third trimester only

5. A 26-year-old at 34 weeks has a routine creatinine of 92 µmol/L, reported as 'normal' by the laboratory (non-pregnant range 45–90). BP 150/98 mmHg. What is the correct interpretation?

A Normal; no further action
B Low; suggests malnutrition
C Expected, because creatinine rises in pregnancy
D Irrelevant to her blood pressure
E Abnormally high for pregnancy, suggesting renal involvement (e.g. pre-eclampsia); she needs further assessment

6. A woman at 36 weeks becomes dizzy, pale and sweaty while lying flat on her back for an ultrasound. BP drops from 118/74 to 86/50 mmHg and the fetal heart slows. What is the FIRST action?

A Give IV adrenaline
B Start IV fluids and call the anaesthetist
C Turn her onto her left side (or manually displace the uterus to the left)
D Perform emergency caesarean section
E Give oxygen by face mask and continue scanning

7. A 25-year-old G1 at 30 weeks has Hb 10.7 g/dL, MCV 88 fL and no symptoms. Her booking Hb at 10 weeks was 12.8 g/dL. According to the WHO/Uganda threshold, what is the correct approach?

A Classify as mild anaemia (Hb below 11 g/dL); continue daily iron–folate, check adherence, exclude malaria and worms, and recheck
B Normal physiological anaemia; no action or supplements needed
C Moderate anaemia requiring admission
D Transfuse 1 unit
E Stop iron because the MCV is normal

8. A woman on postnatal day 5 after caesarean section presents with a painful, swollen left calf. Which physiological change of pregnancy contributes MOST to her risk?

A Increased GFR
B Respiratory alkalosis
C Reduced plasma volume
D Increased clotting factors (fibrinogen, VII, VIII, X), reduced protein S and reduced fibrinolysis
E Increased insulin resistance

9. A 28-year-old at 12 weeks presents with heavy vaginal bleeding and severe vomiting. The uterus is the size of 20 weeks. BP 150/100 mmHg. No fetal heart is detected by Doppler. What is the MOST likely diagnosis?

A Twin pregnancy with threatened miscarriage
B Complete hydatidiform mole
C Uterine fibroid in pregnancy
D Pre-eclampsia with a normal pregnancy
E Incomplete miscarriage

10. A 22-year-old woman reports amenorrhoea for 10 weeks, but her urine pregnancy test is negative. She had a DMPA injection 14 weeks ago. What is the BEST next step?

A Diagnose missed miscarriage
B Refer for hysteroscopy
C Start combined oral contraceptive to induce a bleed
D Order serum prolactin and FSH urgently
E Explain that DMPA commonly causes amenorrhoea, confirm the result (first-morning urine or serum hCG), and give her next DMPA dose

11. A 35-year-old G3P2 at 30 weeks with known rheumatic heart disease presents with orthopnoea, RR 28/min, SpO_2 90% and bibasal crepitations. Why is she decompensating now?

A Plasma volume is falling at this stage

B SVR peaks at 30 weeks

C Cardiac output and blood volume are near their peak, so a stenotic valve cannot cope with the increased flow and tachycardia

D Respiratory alkalosis impairs myocardial contractility

E The uterus stops compressing the IVC

12. A primigravida at 26 weeks has 2+ glycosuria on a routine urine dipstick. She has no symptoms. Her BMI is 24 and there is no family history of diabetes. What is the MOST appropriate step?

A Arrange a blood glucose test (FBS/RBS, or a 75 g OGTT at 24–28 weeks)

B Diagnose GDM and start insulin

C Ignore it; glycosuria in pregnancy is always physiological

D Start metformin

E Restrict all carbohydrates

13. Twenty minutes after a vaginal delivery, a woman who has lost an estimated 1,000 mL of blood has HR 118 bpm and BP 108/70 mmHg. The intern says she is 'stable' because her BP is normal. What is the correct response?

A Agree; observe hourly

B Give oral iron and review tomorrow

C She is compensating; shock index is about 1.1, so treat as significant PPH now (call for help, IV access, fluids, uterotonics, find the cause)

D Perform a CT scan

E Give furosemide for tachycardia

14. A woman at 34 weeks has itching of her palms and soles without a rash. LFTs: ALP 4 times the non-pregnant upper limit, ALT 180 U/L (normal below 35). Which statement is CORRECT?

A Both results are physiological

B Raised ALP proves liver disease

C ALT is always raised in the third trimester

D The ALP is not diagnostic of liver disease in pregnancy, but a raised ALT is abnormal and, with itching, suggests intrahepatic cholestasis of pregnancy

E She needs a liver biopsy

15. A 19-year-old at 16 weeks has had pyelonephritis (fever, loin pain, pyuria). Which physiological change BEST explains her susceptibility?

A Increased bladder tone

B Progesterone-induced dilatation of the ureters and renal pelvis, especially on the right, causing urinary stasis

C Decreased GFR

D Acidic urine from respiratory alkalosis

E Reduced glycosuria

Answers & explanations

1. B · Pelvic ultrasound (preferably transvaginal) to locate the pregnancy

A positive hCG with **pain and bleeding** means **ectopic until proven otherwise**. She is haemodynamically stable, so the next step is an **ultrasound** to find the pregnancy (intrauterine sac, adnexal mass, free fluid). A misses a potential ectopic. C treats a miscarriage that has not been confirmed. D: doxycycline is contraindicated in pregnancy, and PID is unlikely with a positive hCG. E: laparotomy is for an unstable woman or a confirmed ruptured ectopic.

2. D · 27 October 2025

Naegele: **+7 days** gives 27 January; **−3 months** gives 27 October 2024; **+1 year** gives **27 October 2025**. Option B forgets to add 7 days; A subtracts 7 days instead; C and E are arithmetic errors.

3. A · Physiological dyspnoea of pregnancy from progesterone-driven hyperventilation

Gradual breathlessness with a **normal RR, SpO_2 and HR** and a **soft ejection systolic flow murmur** fits **physiological dyspnoea**: progesterone raises tidal volume and minute ventilation. B: mitral stenosis gives a **diastolic** murmur, often with orthopnoea or crepitations. C: PE gives tachypnoea, tachycardia and hypoxia. D: peripartum cardiomyopathy occurs late in pregnancy or postpartum, with signs of heart failure. E: severe anaemia would give tachycardia and pallor.

4. C · Raised for pregnancy, suggesting respiratory fatigue and impending failure

Normal $PaCO_2$ in pregnancy is **about 4 kPa (30 mmHg)** because of progesterone-driven hyperventilation. A $PaCO_2$ of 5.4 kPa is **CO_2 retention** in a pregnant woman having an asthma attack, which means she is **tiring**. Escalate (senior, HDU/ICU, consider ventilation). A and E apply non-pregnant ranges. B is wrong because 5.4 is not low. D: pH 7.38 with a high CO_2 is not metabolic alkalosis.

5. E · Abnormally high for pregnancy, suggesting renal involvement (e.g. pre-eclampsia); she needs further assessment

GFR rises about 50% in pregnancy, so creatinine **falls** (normal about 35–70 µmol/L). A value of **92 µmol/L is abnormal**, and with hypertension it suggests **pre-eclampsia with renal involvement**. A and C apply non-pregnant logic. B is wrong. D ignores the clinical picture.

6. C · Turn her onto her left side (or manually displace the uterus to the left)

This is **supine hypotensive syndrome** from **aortocaval compression**. Relieving the compression with a **left lateral position** or uterine displacement restores venous return within seconds. A, B and D are unnecessary once the cause is removed. E does not treat the cause.

7. A · Classify as mild anaemia (Hb below 11 g/dL); continue daily iron–folate, check adherence, exclude malaria and worms, and recheck

Haemodilution explains some of the fall, but WHO defines anaemia in pregnancy as **Hb below 11 g/dL**. At **10.7** she has **mild anaemia**: ensure iron–folate adherence, deworm, exclude malaria, and recheck. A normal MCV does not exclude early iron deficiency (E). B leaves anaemia untreated. C and D overtreat.

8. D · Increased clotting factors (fibrinogen, VII, VIII, X), reduced protein S and reduced fibrinolysis

Pregnancy is **hypercoagulable**, with raised procoagulants, reduced protein S and reduced fibrinolysis. Combined with **stasis** (venous compression, immobility) and **vessel injury** (CS), this completes Virchow's triad. The risk is highest **postpartum**, especially after CS. A, B and E do not drive thrombosis. C is false: plasma volume **rises**.

9. B · Complete hydatidiform mole

Uterus **larger than dates**, **hyperemesis**, **bleeding**, **hypertension before 20 weeks** and **no fetal heart** is the classic picture of a **complete molar pregnancy** (very high hCG). A: twins would usually have fetal hearts. C: a fibroid does not cause early hypertension. D: pre-eclampsia before 20 weeks suggests a mole. E: the uterus is usually smaller than dates.

10. E · Explain that DMPA commonly causes amenorrhoea, confirm the result (first-morning urine or serum hCG), and give her next DMPA dose

DMPA causes amenorrhoea in many users. A negative urine hCG is reassuring; confirm with a concentrated sample or serum hCG, then continue contraception (she is within the 4-week grace period for her 13-week injection). A: there is no evidence of pregnancy. B, C and D are unnecessary.

11. C · Cardiac output and blood volume are near their peak, so a stenotic valve cannot cope with the increased flow and tachycardia

Cardiac output rises 30–50% and blood volume about 50% by **28–32 weeks**. In **mitral stenosis**, the increased flow plus a higher heart rate (shorter diastolic filling time) raises left atrial pressure and causes **pulmonary oedema**. The next danger point is **immediately after delivery** (autotransfusion). A and B are false (plasma volume rises and SVR falls). D is not the mechanism. E is false.

12. A · Arrange a blood glucose test (FBS/RBS, or a 75 g OGTT at 24–28 weeks)

Increased GFR exceeds the tubular reabsorption of glucose, so **glycosuria is common** in pregnancy. It is **not diagnostic** of diabetes, but persistent or heavy glycosuria is an indication for a **blood test** (she is in the 24–28-week window for an OGTT). B and D treat without a diagnosis. C is dangerous because true GDM would be missed. E is inappropriate.

13. C · She is compensating; shock index is about 1.1, so treat as significant PPH now (call for help, IV access, fluids, uterotonics, find the cause)

Expanded blood volume allows a pregnant woman to lose **up to about 30%** before BP falls. **Tachycardia with a shock index (HR/SBP) of about 1.1** signals significant blood loss. **Hypotension is a late sign.** Manage actively now. A and B are dangerous delays. D is irrelevant. E would worsen hypovolaemia.

14. D · The ALP is not diagnostic of liver disease in pregnancy, but a raised ALT is abnormal and, with itching, suggests intrahepatic cholestasis of pregnancy

ALP rises 2–4 times physiologically (placental isoenzyme), so it cannot be used alone. **Transaminases do not rise in normal pregnancy.** With palmar and plantar pruritus this suggests **obstetric cholestasis** (check bile acids). Also consider HELLP and hepatitis. A, B and C are false. E is not indicated.

15. B · Progesterone-induced dilatation of the ureters and renal pelvis, especially on the right, causing urinary stasis

Progesterone relaxes ureteric smooth muscle, and the gravid uterus compresses the ureters at the pelvic brim (the right is more affected because of dextrorotation). The resulting **hydroureter causes stasis**, so **asymptomatic bacteriuria ascends to pyelonephritis**. A: bladder tone **falls**. C: GFR **rises**. D: urine is not made more acidic by respiratory alkalosis in a way that causes infection. E: glycosuria **increases** and feeds bacteria.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Is she pregnant, and where?

Atim, a 23-year-old P1, attends Mengo casualty with 7 weeks of amenorrhoea and 2 days of lower abdominal pain and brownish spotting. She had an IUD inserted 18 months ago.

1. What history do you take, and why does each part matter?

> **LNMP and cycle** (dating); **nature and site of pain**, **shoulder-tip pain**, **fainting** (haemoperitoneum); amount of bleeding and passage of tissue (miscarriage); **IUD** (reduces pregnancy overall, but if pregnancy occurs, the **risk of ectopic is higher**); previous PID or STI; previous ectopic or tubal surgery; symptoms of pregnancy.

2. Examination: pulse 96 bpm, BP 110/72 mmHg, mild pallor, tender left iliac fossa without guarding. Speculum: IUD strings visible, closed os, minimal old blood. What test first?

> **Urine hCG** (it is positive). Then **transvaginal/pelvic ultrasound**, with a **quantitative serum β-hCG** if available. Also **Hb** and **blood group and cross-match**. Avoid vigorous bimanual examination (risk of rupture).

3. The scan shows an empty uterus with an IUD in situ, a 2.5 cm left adnexal mass, and a small amount of free fluid. Serum β-hCG is 2,800 IU/L. Interpret this.

> An hCG **above the discriminatory zone** (about 1,500–3,500 IU/L) with **no intrauterine sac** and an **adnexal mass plus free fluid** means **left tubal ectopic pregnancy**, probably unruptured or leaking. Differentials (less likely): complete miscarriage, very early IUP (unlikely at this hCG).

4. What is the management at a Ugandan referral hospital?

> She is stable but has free fluid and pain, so **surgical management** (laparoscopic or open **salpingectomy**) is most common and safest in Uganda. Keep her nil by mouth, secure IV access, and cross-match. Methotrexate is an option only if she is stable, hCG is low (below about 1,500–5,000), there is no fetal heart, the mass is small, and reliable follow-up is possible. Remove the IUD. Give **anti-D** if Rh-negative.

5. What counselling and follow-up are needed?

> Explain the diagnosis and the fertility implications (a remaining tube still allows pregnancy). Recurrence risk is about 10–15%, so she should have an **early scan in any future pregnancy**. Offer contraception. Check the histology.

Take-away: A positive hCG plus pain or bleeding is an ectopic until proven otherwise. Correlate the hCG with the ultrasound findings using the discriminatory zone.

Case 2: When physiology meets heart disease

Nansubuga, a 29-year-old G3P2 at 30 weeks, is brought to a regional referral hospital with 3 days of worsening breathlessness, cough and inability to lie flat. She was told as a teenager that she had a 'heart problem' but never followed up.

1. What history do you take?

> Onset and severity of **breathlessness (NYHA class)**, **orthopnoea/PND**, palpitations, haemoptysis, chest pain, leg swelling; rheumatic fever and sore throats in childhood; previous pregnancies (did she decompensate?); penicillin prophylaxis; fever (infective endocarditis, chest infection); anaemia symptoms.

2. Examination: HR 124 bpm irregular, RR 30/min, SpO_2 88%, BP 100/70 mmHg, a mid-diastolic rumble at the apex, loud S1, bibasal crepitations. How do you distinguish this from normal pregnancy?

> In normal pregnancy RR is **not raised**, SpO_2 is normal, HR rises only by 10–20 bpm, and murmurs are **soft and systolic**. Here there is **tachypnoea, hypoxia, an irregular tachycardia (likely AF), a diastolic murmur and crepitations**, which means **pulmonary oedema from mitral stenosis** (rheumatic). Pregnancy has raised cardiac output and HR, shortening diastolic filling.

3. What are your immediate management steps?

> **Sit her up**, **oxygen** to SpO_2 of 95% or more, IV access, **furosemide 40 mg IV** (repeat as needed), and **rate control** (e.g. a beta-blocker such as metoprolol or propranolol; digoxin if AF). **Anticoagulate** (LMWH) if in AF. ECG, echocardiography, FBC, electrolytes. Left lateral tilt. Involve the **physician, obstetrician and anaesthetist**. Watch for fluid overload: avoid large IV fluid volumes.

4. What about delivery planning?

> Aim for a **vaginal delivery** if she is stable at term (CS for obstetric reasons), with **good analgesia (epidural** where available), a **shortened second stage** (assisted delivery), and **avoid supine positioning**. **Avoid ergometrine**; give **oxytocin slowly** (10 IU IM or a slow infusion rather than a bolus). Give **furosemide after delivery** because of autotransfusion. Consider balloon mitral valvotomy in pregnancy if she is refractory (specialist centre).

5. When is she at highest risk, and what follow-up does she need?

> Highest risk at **28–32 weeks**, in **labour**, and **immediately postpartum** (autotransfusion of about 500 mL and relief of IVC compression). Follow-up: cardiology, **benzathine penicillin secondary prophylaxis**, valve intervention before any future pregnancy, and effective contraception (a progestogen implant or IUD; avoid oestrogen).

Take-away: Normal pregnancy never causes tachypnoea, hypoxia or a diastolic murmur. The rise in cardiac output unmasks rheumatic valve disease at 28–32 weeks and just after delivery.

Case 3: The 'stable' bleeding mother

Aber, a 31-year-old P4, delivered at a HC IV 30 minutes ago. The midwife reports that she 'bled a bit' but her BP is 104/68 mmHg, so she is 'stable'. Her pulse is 122 bpm, she is restless and her pads are soaked.

1. Why can her BP be normal despite significant blood loss?

> Pregnancy expands blood volume by about 1.2–1.5 L and she is young with a strong sympathetic response. She can lose **up to about 30% (1–1.5 L)** before BP falls. **Tachycardia, restlessness and a high shock index (122/104 ≈ 1.2)** are the early signs; hypotension is a **late, pre-arrest** sign.

2. What are your immediate steps?

> **Call for help**; ABC; **two large-bore IV cannulae**; take blood for **Hb, grouping and cross-match, and a bedside clotting test**; **rub up the uterus**; start **oxytocin 10 IU IV/IM then an infusion** (e.g. 20–40 IU in 1 L normal saline); **tranexamic acid 1 g IV** over 10 minutes (within 3 hours of birth); crystalloid; **empty the bladder**; find the cause (the **4 Ts**: tone, trauma, tissue, thrombin).

3. Which physiological changes help her survive, and which might betray her?

> **Help:** increased blood volume, **hypercoagulability** (high fibrinogen), and uterine contraction ('living ligatures'). **Betray:** uterine blood flow of about **500–750 mL/min** means she can exsanguinate within minutes; the compensation masks early shock; and if fibrinogen falls below 2 g/L, bleeding worsens (consumption).

4. Her uterus is atonic despite oxytocin. What next at a HC IV?

> Bimanual uterine compression; **misoprostol 800 µg sublingually**; **ergometrine 0.2 mg IM** if she is **not hypertensive**; check the placenta for completeness and the genital tract for tears; **uterine balloon tamponade** (a condom catheter); transfuse. If bleeding continues, go to theatre (B-Lynch suture, uterine artery ligation, hysterectomy) or **refer with a non-pneumatic anti-shock garment** if not available.

5. What follow-up does she need?

> Monitor vital signs, urine output (at least 30 mL/h) and Hb. Give **iron** (or transfuse if Hb is below 7 g/dL or she is symptomatic). Watch for **Sheehan's syndrome** (failure of lactation, amenorrhoea). Plan a hospital delivery with active management in any future pregnancy, and offer family planning.

Take-away: The expanded blood volume of pregnancy hides early shock. Treat tachycardia and a shock index of 0.9 or more as haemorrhage and act before the BP falls.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.